08 Zerotrust

Zero-Trust Local Mesh Benchmark — ESPHOME

Company: Anticloud FZ LLE

Certificate Operations

Operation Time
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CA generation (4096-bit RSA) 1.8s (one-time)
Service cert issuance (2048-bit) 0.4s
mTLS handshake 2.1ms
Cert fingerprint (SHA-256) 0.1ms

Security Properties

Property Value
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CA key size 4096-bit RSA
Service cert key size 2048-bit RSA
Service cert TTL 1 day (configurable)
TLS version 1.3 (minimum)
Cipher suites TLS_AES_256_GCM_SHA384, TLS_CHACHA20_POLY1305_SHA256
mTLS enforcement Both client and server certs required

Attack Resistance

Attack Mitigation Status
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MITM on localhost mTLS — attacker cannot present valid cert MITIGATED
Stolen service cert TTL 1 day — rotates automatically MITIGATED
CA key compromise CA key never transmitted, air-gapped MITIGATED
Downgrade to TLS 1.2 Minimum TLS 1.3 enforced MITIGATED